CHAPTER 7

Improving Query Performance

In this chapter you'll analyze SQL queries running on PostgreSQL, developing a richer understanding of their characteristics as they relate to performance.

To do that you'll learn how to generate and read query execution plans, and use statistics that the database continually collects. Once you understand which portions of a query are most costly, you can begin to use tactics that reduce the costly portions which can improve the overall scalability of your server instance.

Query optimization is a complex subject, with entire books dedicated to covering it. In this chapter you'll get equipped with the basics, learning where to find information and how to interpret it.

Review the following terminology to learn about it for the first time or get refreshed on it. You'll be working with terms like *selectivity*, *cardinality*, a variety of Index types, filtering, sorting, and more in this chapter.

Query Performance Terminology



- Selectivity — How selective a query is
- Cardinality — How many unique values there are
- Sequential Scan — Reading all rows in table
- Index Scan — Fetching values from an Index on a table

To get insights from your query execution, you'll also work with your logs. Before going to PostgreSQL logs, what kind of query information can you get from Active Record logs?